

Ex.no:8	MINI PROJECT – FIFA WORLD CUP 2026 PLAYER PERFORMANCE

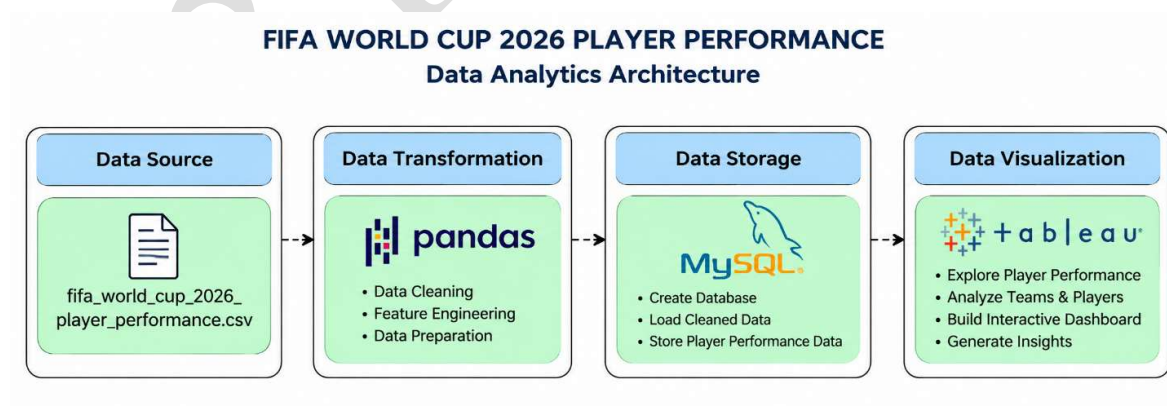
Aim:

To clean, transform, store, and analyze FIFA World Cup 2026 player performance data using Python, MySQL, Power BI and Tableau to derive meaningful insights into player performance, team statistics, goals, assists, ratings and match outcomes.

Algorithm:

1. Load the FIFA World Cup 2026 player performance dataset using Pandas.
2. Clean the data by checking missing values and removing duplicate records.
3. Transform the data by converting match date and creating useful performance measures.
4. Store the cleaned dataset in a MySQL database named `fifa_analytics`.
5. Connect MySQL to Power BI for analysis and reporting.
6. Create DAX measures for total players, goals, assists and average player rating.
7. Create interactive visualizations in Tableau to analyze teams, players, goals, assists and performance.
8. Create a dashboard and generate meaningful insights.
9. End the process.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd

from sqlalchemy import create_engine, text

# Load dataset

df = pd.read_csv("fifa_world_cup_2026_player_performance.csv")

# Create a copy for processing

fifa = df.copy()

# Handle missing values

fifa["player_name"] = fifa["player_name"].fillna("Unknown")
fifa["nationality"] = fifa["nationality"].fillna("Unknown")
fifa["team"] = fifa["team"].fillna("Unknown")
fifa["position"] = fifa["position"].fillna("Unknown")

# Remove duplicate records

fifa.drop_duplicates(inplace=True)

# Convert match date

fifa["match_date"] = pd.to_datetime(fifa["match_date"])

# Create derived columns

fifa["goal_contribution"] = (
    fifa["goals"] + fifa["assists"]
)

fifa["pass_efficiency"] = (
    fifa["successful_passes"] /
    fifa["total_passes"]
)

# Standardize column names

fifa.columns = (
    fifa.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)

print("Original Shape:", df.shape)
print("Cleaned Shape:", fifa.shape)

fifa.head()
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80  
  
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS fifa_analytics;
```

Python/Jupyter:

```
username = "root"  
password = "YOUR_MYSQL_PASSWORD"  
host = "localhost"  
port = 3306  
# Connect to MySQL database  
db_engine = create_engine(  
    f"mysql+pymysql://{username}:{password}@{host}:{port}/fifa_analytics"  
)  
# Load cleaned data into MySQL  
fifa.to_sql(  
    "player_performance",  
    con=db_engine,  
    if_exists="replace",  
    index=False,  
    chunksize=5000  
)  
print("Data loaded successfully into MySQL.")
```

Verify:

```
pd.read_sql(  
    "SELECT COUNT(*) AS total_rows FROM player_performance",  
    db_engine  
)
```

3. Power BI – DAX Measures

Total Players:

```
Total Players =  
DISTINCTCOUNT(player_performance[player_id])
```

Total Goals:

```
Total Goals =  
SUM(player_performance[goals])
```

Total Assists

```
Total Assists =  
SUM(player_performance[assists])
```

Average Player Rating

```
Average Rating =  
AVERAGE(player_performance[player_rating])
```

Average Performance Score

```
Average Performance =  
AVERAGE(player_performance[performance_score])
```

Total Player of Match Awards

```
Player of Match Awards =  
SUM(player_performance[player_of_match_awards])
```

4. Tableau Dashboard

Created visuals in tableau in separate sheets and created a dashboard

Output:

Original Shape: (54600, 75)
Cleaned Shape: (54600, 77)

	player_id	player_name	age	nationality	team	jersey_number	position	height_cm	weight_kg	preferred_foot	...	creativity_score	consistency_score	clutch_performance_score	total_goals_tournament
0	P00055	Rodri Fati	26	Spanish	Spain	3	Goalkeeper	195	75	Left	...	55.9	42.0	51.8	0
1	P00070	Ansu Le Normand	19	Spanish	Spain	18	Midfielder	178	75	Right	...	43.7	31.1	52.7	0
2	P00066	Gavi Ramos	18	Spanish	Spain	14	Midfielder	177	72	Left	...	99.0	83.4	54.8	1
3	P00073	Pedro Cubarsi	20	Spanish	Spain	21	Forward	182	74	Right	...	42.3	40.9	78.5	5
4	P00059	Alvaro Oyarzabal	23	Spanish	Spain	7	Defender	191	81	Left	...	33.5	60.0	56.6	0

5 rows x 77 columns

total_assists_tournament	total_minutes_tournament	player_of_match_aways	tournament_rating	goal_contribution	pass_efficiency
0	242	0	5.8	0	0.576923
3	342	0	5.5	0	0.875000
1	245	0	8.4	1	0.847059
3	422	0	6.7	2	0.631579
0	440	0	5.7	0	0.750000

```
In [15]: username = "root"
password = "" # MySQL root password
host = "localhost"
port = 3306

# Connect to MySQL server and ensure database exists
server_engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}")
with server_engine.connect() as conn:
    conn.execute(text("CREATE DATABASE IF NOT EXISTS fifa_analytics;"))
    conn.commit()

# Connect to MySQL fifa_analytics database
db_engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/fifa_analytics"
)

# Load cleaned data into MySQL
fifa.to_sql(
    "player_performance",
    con=db_engine,
    if_exists="replace",
    index=False,
    chunksize=5000
)

print("Data loaded successfully into MySQL.")
```

Data loaded successfully into MySQL.

```
In [16]: pd.read_sql(
        "SELECT COUNT(*) AS total_rows FROM player_performance",
        db_engine
    )
```

```
Out[16]: total_rows
0      54600
```

Navigator

Display Options

localhost\SQLEXPRESS: fifa_analytics [1]

player_performance

player_performance

player_id	player_name	age	nationality	team	jersey_num
P00055	Rodri Fati	26	Spanish	Spain	
P00070	Ansu Le Normand	19	Spanish	Spain	
P00066	Gavi Ramos	18	Spanish	Spain	
P00073	Pedro Cubarsi	20	Spanish	Spain	
P00059	Alvaro Oyarzabal	23	Spanish	Spain	
P00072	Dani Gavi	21	Spanish	Spain	

The data in the preview has been truncated due to size limits.

FileHomeHelpTable tools

Name player_performance

Manage relationshipsQuick measureNew columnNew tableMark as date table

player_id	player_name	age	nationality	team	jersey_number	position	height_cm	weight_kg	preferred_foot	club_name	market_value_eur	match_id	ma
P00444	Scott McGregor	32	Scottish	Scotland	2	Goalkeeper	188	77	Right	Anderlecht	10797402	M00014	24-0
P00106	Federico Tonalì	38	Italian	Italy	2	Goalkeeper	183	77	Right	Borussia Dortmund	3518278	M00014	24-0
P00105	Gianluigi Chiesa	33	Italian	Italy	1	Goalkeeper	187	75	Right	Borussia Dortmund	2092367	M00014	24-0
P00351	Robert Swierczok	31	Polish	Poland	13	Midfielder	185	82	Right	Benfica	17650534	M00015	12-0
P00355	Grzegorz Skorasz	21	Polish	Poland	17	Midfielder	178	74	Right	Ajax	6697432	M00015	12-0
P00348	Mateusz Frankowski	31	Polish	Poland	10	Defender	184	78	Left	Feyenoord	10570804	M00015	12-0
P00339	Robert Zielinski	28	Polish	Poland	1	Goalkeeper	186	81	Right	AC Milan	1713925	M00015	12-0
P01094	Thomas Ayew	28	Ghanaian	Ghana	2	Goalkeeper	188	81	Right	Wydad Casablanca	17073743	M00015	12-0
P00917	Harry Leckie	32	Australian	Australia	7	Defender	191	80	Right	Erbil SC	98450406	M00024	24-0
P00005	Antoine Koukou	24	French	France	5	Defender	188	79	Right	Borussia Dortmund	21832889	M00024	24-0
P00007	Aurelien Rabiot	25	French	France	7	Defender	184	76	Right	Midtjylland	21875188	M00024	24-0
P01131	Samuel Mouri	22	Cameroonian	Cameroon	13	Midfielder	177	77	Right	Al-Hilal Omdurman	3430408	M00025	29-0
P01122	Samuel Toko-Ekambi	27	Cameroonian	Cameroon	4	Defender	190	81	Right	Al Ahly	3575619	M00025	29-0
P01127	Nicolas Siani	31	Cameroonian	Cameroon	9	Defender	188	80	Left	Kaizer Chiefs	2163073	M00025	29-0
P01138	Moumi Choupo-Moti	30	Cameroonian	Cameroon	20	Midfielder	172	69	Right	JS Kabylie	17926924	M00025	29-0
P01119	Samuel Choupo-Moti	26	Cameroonian	Cameroon	1	Goalkeeper	189	78	Right	ES Tunis	74850274	M00025	29-0
P00891	Yasser Al-Owais	22	Saudi	Saudi Arab	7	Defender	184	77	Right	Al-Shorta	2896502	M00025	29-0
P00887	Saleh Al-Otaibi	34	Saudi	Saudi Arab	3	Goalkeeper	192	78	Right	Pakhtakor	3736086	M00025	29-0
P00349	Jakub Helik	18	Polish	Poland	11	Defender	179	73	Right	Copenhagen	3656412	M00048	19-0
P00348	Mateusz Frankowski	31	Polish	Poland	10	Defender	184	78	Left	Feyenoord	10570804	M00048	19-0
P00342	Robert Krychowiak	27	Polish	Poland	4	Defender	182	83	Right	Malmö FF	7306230	M00048	19-0
P00339	Robert Zielinski	28	Polish	Poland	1	Goalkeeper	186	81	Right	AC Milan	1713925	M00048	19-0
P00927	Ajdin Jones	27	Australian	Australia	17	Midfielder	179	79	Left	Yokohama F. Marinos	1579010	M00048	19-0
P00912	Aaron Souttar	28	Australian	Australia	2	Goalkeeper	182	77	Left	Ulsan HD	8434271	M00048	19-0
P00098	John Bellingham	29	English	England	20	Midfielder	185	78	Right	Marseille	12779416	M00065	28-0
P00084	Phil Walker	30	English	England	6	Defender	184	82	Left	Barcelona	2890294	M00065	28-0
P00083	Jude Grealish	23	English	England	5	Defender	181	68	Left	FC Basel	42246372	M00065	28-0

Table: player_performance (54,600 rows)

ResultsResult 1 of 1Copy

	[Total Players]	[Total Matches]	[Total Goals]	[Total Assists]	[Average Rating]	[Average Performan...]
1	1248	1050	3024	2859	3.63	36.83

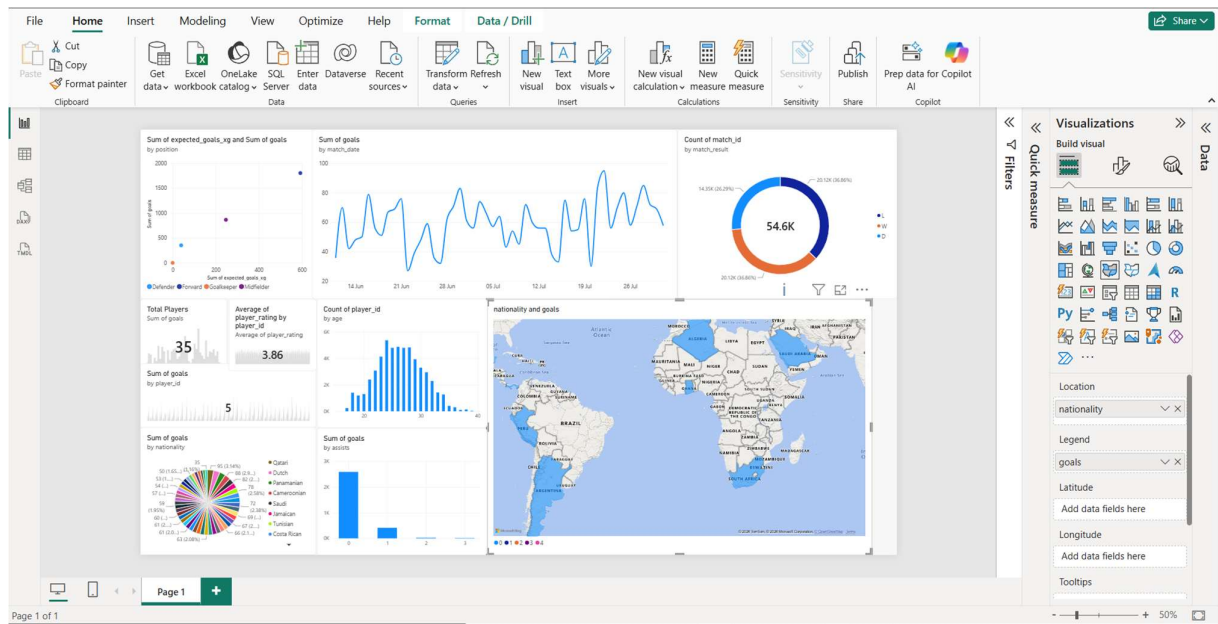


Tableau Public - Book1

File Data Window Help

Connections

- Live SQL Server (Dinesh\SQL...

Table

- Live SQL Server Data

New Table Extension

Live SQL Server Data

Name: Live SQL Server Data

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote ...
Abc	Player Id	Live SQL Server Data	player_id
Abc	Player Name	Live SQL Server Data	player_na...
Abc	Nationality	Live SQL Server Data	nationality

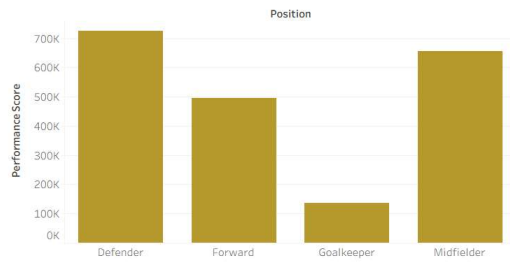
Update Now

Update Automatically

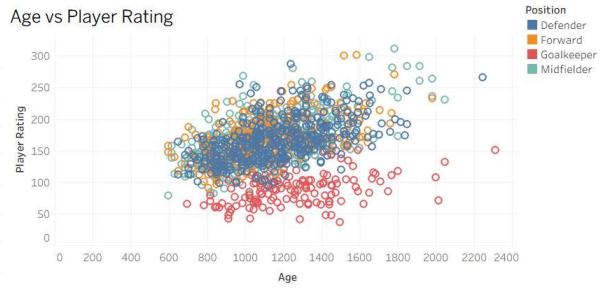
Data Source: Sheet 1

B Dinesh

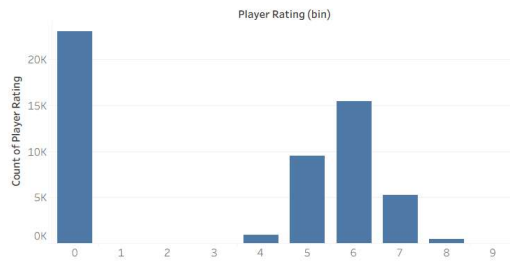
Performance by Position



Age vs Player Rating



Player Rating Distribution



Market Value vs Player Rating



Result: